MILITARY SPECIFICATION

COMPUTER, BALLISTICS: XM19

1. SCOPE

1.1 This specification covers one type analog computing instrumentation known as Computer, Ballistics: XM19 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of.
MIL-P-14232	Parts, Equipment and Tools for Ordnance Material; Packaging of.
MIL-I-45607	Inspection Equipment, Supply and Maintenance of.
MIL-B-60124	Ballistics Computer Subassembly: 8635412 (Cant Unit).
MIL-B-60597	Ballistics Computer Subassembly: 10543800 (Computer Unit)
MIL-B-60598	Ballistics Computer Subassembly: 10543700 (Commander's Control Unit).
MIL-B-60599	Ballistics Computer Subassembly: 10543600 (Gunner's Control Unit).

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	Quality Assurance Terms and Definitions.

DRAWINGS

U. S. Army, Frankford Arsenal

F10543900	Computer, Ballistics: XM19
IEL 10543900	List of Inspection Documents for Computer: XM19

PACKAGING DATA SHEET

MIL-P-14232/P10543900	Packaging of Computer, Ballistics: XM19

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication.- The computer shall be manufactured in accordance with Drawing F10543900 and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance and permanence to insure inherent quality and performance capabilities.

3.2 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Part identification and marking.
(e) Workmanship.

3.3 First article testing.- Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Subassemblies.- Each Computer Ballistics XM19 submitted in conformance with this specification shall be composed of subassemblies, herein referenced as units, which will meet the requirements specified herein; or shall have complied with all their respective requirements as specified in the following specifications.

Ballistics Computer Subassembly: 8635412 (Cant Unit) MIL-B-60124.
Ballistics Computer Subassembly: 10543800 (Computer Unit) MIL-B-60597
Ballistics Computer Subassembly: 10543600 (Gunner's Control Unit) MIL-B-60599
Ballistics Computer Subassembly: 10543700 (Commander's Control Unit) MIL-B-60598

3.5 Environmental.

3.5.1 Storage temperatures.- There shall be no evidence of physical failures such as glass breakage or casting failures due to storage temperatures ranging from -65° to 160° Fahrenheit.

3.5.2 Operating temperatures.- The control unit gunner's, control unit commander's, and computer unit shall meet all the requirements of 3.6 through 3.7.19 inclusive while exposed and thermally stabilized at ambient temperatures of -40° and +150° Fahrenheit (F). Upon return to standard ambient temperature, they shall meet the requirements of 3.5.3.

3.5.3 Vibration.- The slippable scales on the boresight knobs of the commander's and gunner's control units shall be movable under normal finger pressure, but shall not move under vibration. All units shall meet the requirements of 3.6 through 3.7.19 after being vibrated in a vertical plane for five minutes ± 15 seconds at an amplitude of 1/16 inch (1/8 inch total excursion) at a frequency of 30 cycles per second (cps).

3.5.4 Water resistance (sealing).- The interiors of the units shall show no evidence of water penetration or other damage after being subjected to a simulated rain fall as specified herein.

3.6 Electrical.

3.6.1 Continuity.-The continuity of the units and computer system (see 3.4). shall conform to their respective Electrical Schematic. Drawings: F10543841, F10543839, and F10543833.

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3.7 Performance.

3.7.1 Computer power inputs.- All power inputs shall be supplied simultaneously and the computer system shall be operable when the specific electrical power is applied to each cable assembly and its specific connector as outlined below:

(a) Cable Assembly 10543810, Connector MS3116F14-15S.- Supply 18 to 30 volts direct (dc) current power (to be referred to as dc input power and adjusted to the level indicated by the specific test) between contacts P and E of Connector MS3116F14-15S. Positive connection to contact P.

(b) Cable Assembly 10543816, Connector MS3116F14-15S.- Supply 18 to 30 volts dc power, same as in (a) above, between contacts P and A of Connector MS3116F14-15S. Positive connection to contact P.

(c) Cable Assembly 10543815, Connector MS3106R22-15S.- Supply 115 volts rms ± 5%, 400 cps ± 1%, power between contacts A and F of Connector MS3106R22-15S. Harmonic distortion shall not exceed 4%. Variation in harmonic distortion shall not exceed 1.5% during all computer tests. Both ac power lines shall be isolated from all computer circuitry except as indicated by System Schematic F10543874.

3.7.2 Computer connection.- The computer shall be connected in accordance with Drawing F10543900 except for the following simulated conditions:

(a) Computer output servos.- Separate servo output mechanisms shall be used to simulate the periscope reticle drives used in the Gunner's Periscope XM50, 10539750 and Commander's Periscope XM51, 10540250. These servo output mechanisms shall be indicating devices electrically and mechanically appropriate to display simultaneously the gunner's and commander's elevation and deflection corrections solved by the computer.

(b) Vehicle turret to cupola slip ring.- A junction box wired to simulate the slip ring is to be used between Cable Assembly 10543809 and Cable Assembly 10543808.

(c) Vehicle control unit (gunner's).- A simulated unit with controls to represent the gunner's left and right lead lock switches, gun EFC switch and wired to deliver the power specified in 3.7.1 (b).

(d) Vehicle control unit (commander's).- A simulated unit with controls to represent the commander's left and right lead lock switches, cupola co-axial, main gun switch and wired to deliver the power specified in 3.7.1 (a).

(e) Inverter.- The power requirements specified in 3.7.1 (c) shall be used to simulate the inverter.

(f) Laser rangefinder.- A simulated unit with controls to generate digitally coded range signals, laser ranging in process signal, and laser ranging complete signal, shall be connected to the laser rangefinder receptacle of the gunner's control unit.

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3.7.3 Periscope and wiper power switch, control unit, gunner's.- Operation of the periscope and wiper power switch from the center "OFF" to the "ON PERISCOPE" position shall apply dc input power between contacts B and C of Connector 10543875 on Cable Assembly 10543817. Contact B shall test positive with respect to contact C. Operation of the periscope and wiper switch from the "ON PERISCOPE" to the "ON PERISCOPE AND WIPER" position shall apply dc input power between contacts B and C of Connector 10543875 on Cable Assembly 10543817 and contacts C and A of Connector MS3116F8-4S on Cable Assembly 10543817. Contacts B and C shall test positive with respect to contacts C and A respectively, at their corresponding connectors. DC input power shall be applied continuously between contacts B and A of Connector MS3116F8-4S on Cable Assembly 10543817 in all three positions of the periscope and wiper power switch. Contact B shall test positive with respect to contact A.

3.7.4 Periscope power switch, control unit, commander's.- Operation of the periscope power switch from the "OFF" to the "ON" position shall apply dc input power between contacts D and B of Connector 10543875 on Cable Assembly 10543810. Contact D shall test positive with respect to contact B.

3.7.5 Periscope - Wiper Power Switch, control unit, commander's.- Operation of the wiper power switch from the "OFF" to the "ON" position shall apply dc input power between contacts C and A of Connector MS3116F8-4S on Cable Assembly 10543810. Contact C shall test positive with respect to contact A. DC input power shall be continuously present between contacts B and A of Connector MS3116F8-4S on Cable Assembly 10543810 when the wiper switch is in either "ON" or "OFF" position. Contact B shall test positive with respect to contact A.

3.7.6 Computer power switches, gunner's and commander's.- With the computer power input connected as specified in 3.7.1, dc computer power shall be applied when both the gunner's and commander's computer power switches are in the "ON" position. Computer power shall be removed when either the gunner's or commander's computer power switch is moved to the "OFF" position.

3.7.7 Ammunition selection and indication, commander's and gunner's units.- The following requirements shall be met with computer power applied.

3.7.7.1 Color coding.- The selection of ammunitions shall insure that the proper color of the indicator is compatible with the color of the selected type ammunition as depicted on Drawings F10543700, sheet 1 and F10543600, sheet 1.

3.7.7.2 Ammunition selector switch engagement and indication.- The operation of each ammunition selector switch shall cause the computer to change ammunition types and so indicate by an increase in the light intensity of the selected switch that the computer has changed to that ammunition type. The type of ammunition selected shall simultaneously appear on both the commander's and gunner's control units.

3.7.7.3 Input power operation.- Ammunition selection shall function properly for all types under a power input source of 18.0, 24.0 and 30.0 volts dc.

4.

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3.7.8 Boresight - Normal switch operation.- With computer power applied, 1000 meters range set into gunner's control unit, the commander's controls set to main gun, and the main gun boresight knobs at the gunner's and commander's control units set to their electrical center, (scale slipped to the scribe line and knob turned to the zero setting) operation of the boresight normal switch to the boresight position shall cause the computer output servos to zero within 0.05 mils. Operation of the boresight normal switch to the normal position shall restore normal computer solutions to the computer output servos.

3.7.9 Boresight knobs.- The boresight knobs shall function to control the output of the computer output servos within 0.075 mils of the boresight scale markings. Each pull-to-turn boresight knob shall engage the locking gears upon release.

3.7.10 Jump knobs.- The jump knobs shall function to control the output of the computer output servos within 0.075 mil of the jump scale markings. Each pull-to-turn jump knob shall engage the locking gears upon release.

3.7.11 Override controls.- The gunner's and commander's override controls shall function to transfer control to and from each station. With control at the gunner's station (green control illuminated), range and wind switch in manual position, the gunner's range knob, wind knob, cant on-off switch and lead switches shall function effectively to control computer inputs. Under the same conditions, the commander's range knob, wind knob, cant on-off switch and lead switches shall be ineffective in controlling the inputs to the computer and the red (no control) light shall be illuminated on the commander's control unit. Upon transfer of control, the commander's controls shall become effective and the gunner's ineffective. The control-no control lights shall indicate the transfer of control.

3.7.12 Cupola, co-axial and main gun modes, commander's station.- With the commander's simulated control in the cupola position, the commander's computer output servos shall be controlled by the cupola boresight knobs. With the commander's simulated control in the co-ax position, the commander's computer output servos shall be controlled by the commander's main gun boresight knobs and main gun parallax corrections. With the commander's simulated control in the main gun position, the commander's computer output servo shall be controlled by the commander's main gun boresight knobs and the main gun computer solution.

3.7.13 Range input, commander's unit.- With the range switch on the gunner's control unit positioned in "MANUAL" and the control indicator light indicating control at the commander's unit, any range between 350 and 3950 meters introduced into the computer by the commander's unit range dial shall be reflected on the gunner's unit range counter within $\pm$ 5 meters.

3.7.14 Range input, simulated laser rangefinder.- With the range switch on the gunner's control unit positioned in "AUTOMATIC", the gunner's control unit range counter shall move to 1000 meters $\pm$ 5 meters. Application of the laser ranging in process signal from the laser simulator shall cause the computer to generate boresight solutions. Removal of this signal shall return the computer to normal solutions. The application of the designated digital coded range signals in conjunction with the laser ranging complete signal shall drive the gunner's range counter to the sum of the range signal inputs, in accordance with Table I.

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TABLE I

Digital Input Signal	Gunner's Range Counter Reading
2000	2000 ± 3 meters
1000	1000 ± 3 meters
800	800 ± 3 meters
400	400 ± 3 meters
400, 200	600 ± 6 meters
400, 100	500 ± 6 meters
200, 100, 80	380 ± 6 meters
200, 100, 80, 40	420 ± 6 meters
200, 100, 80, 40, 20	440 ± 6 meters
200, 100, 80, 40, 20, 10	450 ± 6 meters (Check for 10 meter increment)
200, 100, 80, 40, 20, 10, 5	455 ± 6 meters (Check for 5 meter increment)

3.7.15 Range limits.- The gunner's range mechanism shall be capable of range inputs from 320 meters to 3980 meters. The commander's range dial shall be capable of range inputs from 325 meters to 3975 meters.

3.7.16 EFC, effective full charge count.- The EFC assembly of the computer shall step one EFC count for each actuation of the EFC switch, in the gunner's vehicle box simulation, with the computer on HEAT-M409 ammunition.

3.7.17 Automatic-manual wind switch.- The gunner's or commander's wind inputs (depending on which station has control) shall be effective on computer operation when the wind switch is in the manual position. The manual wind inputs shall be ineffective when the wind switch is in the automatic position.

3.7.18 Ballistics computer solutions.- The ballistics computer shall produce the solutions specified in Tables II, III, IV, V, VI, VII. These solutions shall be measured on the computer output servos as changes in output from the 1000 meter boresight solutions. The boresight knobs must be set to their electrical center. The computer and simulated vehicle related controls shall be operated from the gunner's controls for gunner's solutions and from the commander's controls for commander's solutions.

3.7.19 Illumination.

3.7.19.1 Range and wind indicators.- The gunner's range counter, wind dial and the commander's range and wind dials shall be legible in a darkened area when input power is applied to the computer system.

3.7.19.2 Override control and ammunition selection indicators.- The rheostats at the gunner's and commander's control units shall control the light intensity of the override control and ammunition selection indicators from dim to full bright.

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Table II. Gunner's solutions, zero wind and lead input tolerances all solutions in this table ± 0.2 mils

Target range meters	+15° Cant Deflection Mils	+15° Cant Elevation Mils	0° Cant Deflection Mils	0° Cant Elevation Mils	-15° Cant Deflection Mils	-15° Cant Elevation Mils
350	0.229	4.098	1.307	4.268	2.386	4.170
500	-1.016	5.795	0.563	6.046	2.142	5.903
750	-2.462	8.904	-0.042	9.298	2.378	9.073
1000	-3.732	12.355	-0.437	12.904	2.857	12.585
1200	-4.805	15.374	-0.653	16.061	3.497	15.665
1500	-6.472	20.437	-0.978	21.353	4.512	20.824
1750	-8.024	25.195	-1.231	26.327	5.557	25.674
2000	-9.728	30.548	-1.493	31.920	6.735	31.128
2250	-11.652	36.574	-1.773	38.216	8.095	37.270
2500	-13.846	43.345	-2.068	45.289	9.699	44.172
2700	-15.812	49.393	-2.302	51.636	11.197	50.352
2900	-18.045	56.048	-2.553	58.585	12.928	57.142

- Is Reticle to the Right; Gun to the Left

+ Is Reticle to the Left; Gun to the Right

Table III. Commander's solutions, zero wind and lead input tolerances all solutions in this table ± 0.3 mils

Target range meters	+15° Cant Deflection Mils	+15° Cant Elevation Mils	0° Cant Deflection Mils	0° Cant Elevation Mils	-15° Cant Deflection Mils	-15° Cant Elevation Mils
350	-1.225	5.721	-0.146	5.892	0.932	5.794
500	-1.793	6.662	-0.213	6.913	1.366	6.771
750	-2.744	9.219	-0.324	9.613	2.096	9.388
1000	-3.732	12.355	-0.437	12.904	2.857	12.585
1200	-4.685	15.239	-0.533	15.927	3.617	15.531
1500	-6.209	20.143	-0.716	21.059	4.775	20.530
1750	-7.703	24.836	-0.910	25.967	5.878	25.314
2000	-9.348	30.124	-1.114	31.496	7.115	30.704
2250	-11.213	36.085	-1.336	37.728	8.534	36.781
2500	-13.364	42.810	-1.587	44.755	10.180	43.637
2700	-15.316	48.835	-1.806	51.079	11.693	49.795
2900	-17.534	55.477	-2.042	58.015	13.440	56.571

- Is Reticle to the Right; Gun to the Left

+ Is Reticle to the Left; Gun to the Right

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Table IV. Gunner's solutions, zero lead input, tolerances all solutions in this table ± 0.2 mils

Target range meters	0° Cant 20 MPH wind From left Deflection Mils	0° Cant 20 MPH wind From right Deflection Mils	+15° Cant 30 MPH wind From right Deflection Mils	+15° Cant 30 MPH wind From right Elevation Mils	-15° Cant 30 MPH wind From left Deflection Mils	-15° Cant 30 MPH wind From left Elevation Mils
350	0.668	1.947	1.188	4.345	1.426	4.418
500	-0.379	1.505	0.396	6.160	0.729	6.268
750	-1.486	1.402	-0.295	9.465	0.211	9.633
1000	-2.422	1.547	-0.753	13.126	-0.121	13.356
1200	-3.110	1.803	-1.117	16.328	-0.189	16.620
1500	-4.157	2.199	-1.700	21.672	-0.256	22.059
1750	-5.111	2.646	-2.199	26.702	-0.263	27.180
2000	-6.093	3.103	-2.822	32.332	-0.164	32.912
2250	-7.225	3.672	-3.468	38.683	-0.080	39.378
2500	-8.395	4.251	-4.348	45.783	0.210	46.608
2700	-9.290	4.676	-5.323	52.114	0.716	53.071
2900	-10.231	5.115	-6.521	59.020	1.410	60.111

- Is Reticle to the Right; Gun to the Left

+ Is Reticle to the Left; Gun to the Right

Table V. Commander's solutions, zero lead input, tolerances all solutions in this table ± 0.3 mils

Target range meters	0° Cant 20 MPH wind From left Deflection Mils	0° Cant 20 MPH wind From right Deflection Mils	+15° Cant 30 MPH wind From right Deflection Mils	+15° Cant 30 MPH wind From right Elevation Mils	-15° Cant 30 MPH wind From left Deflection Mils	-15° Cant 30 MPH wind From left Elevation Mils
350	-0.786	0.493	-0.265	5.969	-0.027	6.042
500	-1.155	0.728	-0.380	7.027	-0.047	7.136
750	-1.768	1.120	-0.577	9.780	-0.070	9.948
1000	-2.422	1.547	-0.753	13.126	-0.121	13.356
1200	-2.991	1.923	-0.998	16.194	-0.069	16.485
1500	-3.894	2.461	-1.438	21.378	-0.006	21.765
1750	-4.790	2.967	-1.879	26.342	0.058	26.821
2000	-5.714	3.482	-2.443	31.908	0.215	32.488
2250	-6.786	4.110	-3.030	38.194	0.357	38.889
2500	-7.914	4.732	-3.867	45.249	0.690	46.074
2700	-8.793	5.172	-4.827	51.557	1.211	52.515
2900	-9.719	5.626	-6.010	58.450	1.921	59.542

- Is Reticle to the Right; Gun to the Left
+ Is Reticle to the Left; Gun to the Right

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TABLE VI

GUNNER'S AND COMMANDER'S TARGET LEAD DEFLECTION SOLUTIONS, ZERO WIND INPUT, CANT OFF, ELEVATION SOLUTION CHECK NOT REQUIRED FOR LEAD TEST

TARGET RANGE METERS	DEFLECTION SOLUTIONS - MILS TRACKING RATE AND DIRECTION - MILS/SEC						
	28.0 CW	12.0 CW	4.0 CW	0.0	4.0 CCW	12.0 CCW	28.0 CCW
350	C 14.85±1.3	C 6.28±0.8	C 1.99±0.5	*	C -2.29±0.5	C -6.58±0.8	C -15.15±1.3
500	G 22.65±1.9	G10.03±1.0	G 3.72±0.6	*	G -2.60±0.6	G -8.91±1.0	G -21.53±1.9
750	*	C14.14±1.3	C 4.50±0.8	*	C -5.14±0.8	C-14.78±1.3	*
1000	*	G19.61±1.5	G 6.24±0.9	*	G -7.12±0.9	G-20.49±1.5	*
1200	*	C24.15±1.9	C 7.70±1.0	*	C -8.76±1.0	C-25.21±1.9	*
1500	*	G31.54±2.3	G 9.86±1.2	*	G-11.82±1.2	G-33.50±2.3	*
2000	*	*	C14.47±1.8	*	C-16.69±1.8	*	*
2500	*	*	G19.27±2.2	*	G-23.41+2.2	*	*
2900	*	*	G23.73±2.6	G-2.55±1.4	G-28.83±2.6	*	*

- Is Reticle to the Right; Gun to the Left
+ Is Reticle to the Left; Gun to the Right
* No Test Required
C Is Commander's Solution
G Is Gunner's Solution

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Table VII. Commander's solutions, co-ax mode, tolerance for elevation & deflection solutions ± 0.3 mils

Range Meters	Elevation Mils	Deflection Mils
350	1.765 Gun Up	0.0
500	0.943 Gun Up	0.0
1000	0.0	0.0
1500	0.319 Gun Down	0.0
2000	0.460 Gun Down	0.0
3000	0.629 Gun Down	0.0
3500	0.676 Gun Down	0.0

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article testing.- When the contract specifies testing of preproduction or initial production samples, the following paragraphs shall apply as applicable.

4.2.1 Preproduction test.- Proproduction test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, two preproduction samples shall be submitted to the procuring agency for approval of design and construction features. Preproduction samples shall meet all inspection provisions of 4.3, 4.4, and 4.6 as well as any additional tests that may be specified in the contract.

4.2.2 Initial production test.- Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4, and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

4.2.3 First article disapproval.- Failure of any one of a first article sample submitted shall be cause for disapproval by the Government and subsequent action as required by the contract clauses applicable to first article testing.

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4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926, and the Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (LID), IEL 10543900. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105, and the sampling plans specified in Table VIII herein. Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE VIII-CLASSIFICATION OF DEFECTS

CLASS		REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR: AQL 0.65% Defective			
101.	Vibration	3.5.3	4.6.1.1
102.	Computer power inputs	3.7.1	4.6.2
103.	Computer connections	3.7.2	4.6.3
104.	Gunner's periscope & wiper power switch, control unit, gunner's	3.7.3	4.6.4
105.	Commander's periscope power switch, control unit	3.7.4	4.6.5
106.	Commander's periscope - wiper switch, control unit	3.7.5	4.6.6
107.	Computer power switches, gunner's & commander's	3.7.6	4.6.7
108.	Ammunition selection and indication	3.7.7	4.6.8
109.	Boresight - normal switch operation	3.7.8	4.6.9
110.	Boresight knobs	3.7.9	4.6.10
111.	Jump knobs	3.7.10	4.6.11
112.	Override controls	3.7.11	4.6.12
113.	Cupola, co-axial, and main gun modes - commander's station	3.7.12	4.6.13
114.	Range input, commander's unit	3.7.13	4.6.14
115.	Range input, simulated laser rangefinder	3.7.14	4.6.15
116.	Range limits	3.7.15	4.6.16
117.	EFC Count	3.7.16	4.6.17
118.	Automatic-manual wind switch operation	3.7.17	4.6.18
119.	Ballistic computer solutions	3.7.18	4.6.19
120.	Illumination, range & wind indicators	3.7.19.1	4.6.20
121.	Illumination, override control and ammo indicators	3.7.19.2	4.6.21
MINOR: NONE			

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4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and re-submittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General. - One ballistic computer shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table IX.

TABLE IX

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Fabrication	3.1	Applicable drawing - visual
302. General specification	3.2	MIL-F-13926 - visual
303. Continuity	3.6.1	Applicable drawings visual - ohmmeter

4.4.2 Environmental. - Three ballistics computers shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall have met the requirements and tests in Table VIII and shall them meet the requirements and tests in Table X.

TABLE X

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304. Storage temperatures.	3.5.1	4.6.1.2
305. Operating temperatures.	3.5.2	4.6.1.3
306. Water Resistance (sealing)	3.5.4	4.6.1.4

4.4.3 Failure of sample.-Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

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4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents (LID), IEL 10543900.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment.- In conjunction with 4.5.2.2, the standard test equipment listed in Table XI shall be utilized in the performance of the applicable tests as specified in 4.6.

TABLE XI

	NOMENCLATURE	DESCRIPTION
1.	D.C. power supply (simulated vehicle source)	18 to 30 volts maximum ripple, 0.5 volt peak to peak, 20 ampere output.
2.	A.C. power supply (simulated inverter)	115 volts rms ± 5%, 400 cps ± 1%, 4% maximum harmonic distortion, 1.5% max. change in harmonic distortion. Output to the computer to be switched on and off by the computer power signal.
3.	Simulated computer output servos	Electrically and mechanically specified in Figures 1, 2, and 3.
4.	Cant unit positioner	Position cant unit with mounting surface horizontal (0°) and at ± 15° of cant within 0.05°.
5.	Gunner's vehicle control unit (simulated input)	Provides gunner L & R lead lock switch and main gun EFC counting switch per Figure 4. DC power to computer by item 1 above.
6.	Commander's vehicle control unit (simulated input)	Provides commander L & R lead lock switch and cupola, co-ax, and main gun switches per Figure 5. DC power to computer by item 1 above.
7.	Simulated laser rangefinder input & DC power supply	Provide digital range signals, laser ranging "in process" switch and laser ranging "completed" switch. Connected to gunner's computer unit #10543600 per Figure 6.
8.	Simulated cupola slip ring	Junction box or adapter providing electrical connection of Cables 10543808 and 10543809.
9.	Powered rotary table	Rotates computer unit #10543800 in its operating position at adjustable angular velocities from .5 mil/sec. to 65 mil/sec. within an accuracy of 1% of indicated velocity.

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4.6 Test methods and procedures.

4.6.1 Environmental.

4.6.1.1 Vibration.- To determine conformance to 3.5.2, the computer unit, commander's control unit, and gunner's control units, without cable assemblies, shall be mounted separately or collectively on a suitable mounting apparatus and attached to a vibrating machine or fixture (see 4.5.2.2) and vibrated at the amplitude, frequency and time specified. After vibration, the units shall be given a visual-tactile examination for damage and loose parts and scale movement. The units shall then be subjected to the tests specified in 4.6.2 to 4.6.21 inclusive to determine conformance to the respective requirements referenced therein.

4.6.1.2 Storage temperatures.- To determine conformance to 3.5.1, the Computer Ballistics, XM19 shall be tested in accordance with Temperature Cycling Procedure II of MIL-F-13926 for the specified temperatures. At the conclusion of these tests, the units shall show no evidence of damage or deterioration when visually examined.

4.6.1.3 Operating temperatures.- To determine conformance to 3.5.3, Computer Ballistics, XM19 shall be placed in appropriate environmental chambers (see 4.5.2.2). While thermally stabilized at the specified temperatures, the units shall be subjected to the tests specified in 4.6.2 to 4.6.21 inclusive to determine conformance with the requirements referenced therein.

4.6.1.4 Water resistance.- To determine conformance to 3.5.4, Computer Ballistics, XM19 shall be tested in accordance with Rain Procedure I of MIL-F-13926. The Computer Ballistics, XM19 shall show no evidence of water penetration as a result of this test when visually examined.

4.6.2 Computer power inputs.- To detemine conformance to 3.7.1, the AC & DC power supplies of Table XI shall be connected to the XM19 Computer Ballistics at the connectors specified and conforming to Figure 4 and 5.

4.6.3 Computer connections.- To determine conformance to 3.7.2, the connections shall be made to the items specified in accordance with Figures 1 thru 6 and Drawing F10543900 using all the equipment listed on Table XI except the rotary table. AC power shall be nominal. DC power shall be set for 24V unless otherwise specified during subsequent tests. Cant unit positioner shall be set to 0°.

4.6.4 Periscope and wiper power switch, control unit, gunner's.- To determine conformance to 3.7.3, the specified switch shall be actuated to the "ON PERISCOPE" position and the specified contacts shall show DC input power and polarity. The switch shall then be actuated to the "ON PERISCOPE AND WIPER" position. DC input and polarity shall be as specified. Checks shall be made in all three switch positions for continuous DC input and proper polarity for contacts B and A as specified.

4.6.5 Periscope power switch, control unit, commander's.- To determine conformance to 3.7.4, the switch shall be actuated from "OFF" to "ON" and the contacts shall show DC input and polarity as specified.

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4.6.6 Periscope-wiper switch, control unit, commander's.- To determine conformance to 3.7.5, the switch shall be actuated from "OFF" to "ON" and the contacts shall show dc input and polarity as specified. Checks shall also be made in both the "OFF" and "ON" positions for continuous dc input and polarity for contacts B and A as specified.

4.6.7 Computer power switches, gunner's and commander's.- To determine conformance to 3.7.6, both gunner's and commander's power switches shall be actuated to the "ON" position and the control units observed for illumination of ammo indicators and range and windage dials. Actuation of either one of the switches shall cause the lights to go out.

4.6.8 Ammunition selection and indication.- To determine conformance to 3.7.7, with computer power "ON", each ammo button shall be depressed individually and observed for increase in brilliance upon actuation. The dc power supply shall be reset to 18V and the test repeated. This will be done again at 30V.

4.6.9 Boresight - normal switch operation.- To determine conformance to 3.7.8, with controls and settings specified, the boresight - normal switch shall be put in the "BORESIGHT" position and both sets of simulated computer output servo indicators of Table XI shall zero within 0.05 mils. Returning the switch to "NORMAL" shall return the indicators to their original setting.

4.6.10 Boresight knobs.- To determine conformance to 3.7.9, one boresight knob at a time shall be pulled out and turned to three separate readings in each direction of travel. Upon releasing, the knob shall be observed for proper locking engagement of the gears. At each reading, the corresponding computer output servo indicator shall be observed for movement to the new setting within the tolerance specified. All six boresight knobs shall be checked. When checking cupola gun and main gun with the commander's boresight knobs, the commander's vehicle control unit of Table XI shall be set to the proper switch.

4.6.11 Jump knobs.- To determine conformance to 3.7.10, the gunner's ammunition jump knobs shall be tested in the same manner as 4.6.10. All eight jump knobs shall be tested.

4.6.12 Override controls.- To determine conformance to 3.7.11, with gunner's range and wind switches in "MANUAL" and "NO CONTROL" showing on the gunner's control panel, depressing either override button shall cause the green light to illuminate at the gunner's controls while the commander's indicating light goes to red. The gunner's controls specified shall be actuated and their corresponding computer function noted. Depression of the commander's override control button shall cause the commander's "CONTROL" indicating light to illuminate green and the gunner's "NO CONTROL" light to illuminate red. With commander in "CONTROL", actuation of the specified switches and knobs shall cause the computer function desired. When either control unit is in the "NO-CONTROL" mode, actuation of any of the knobs and switches specified in 3.7.11 for that mode shall be ineffective.

4.6.13 Cupola, co-axial, and main gun modes, commander's station.- To determine conformance to 3.7.12, commander's cupola boresight knobs control shall be checked in conjunction with 4.6.10. Commander's co-axial control shall be checked in conjunction with 4.6.19.7 and main gun control in conjunction with 4.6.19.2.

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4.6.14 Range input, commander's unit.- To determine conformance to 3.7.13, with the gunner's range switch on "MANUAL" and the commander's control light "GREEN", the commander's range knob shall be turned to a setting of 350 meters. The gunner's range counter shall read between 345 and 355. Advancing the commander's range dial to the following ranges shall reflect a corresponding reading on the gunner's dial within the tolerance specified: 600, 1000, 1400, 1900, 2400, 3000, 3500, and 3950.

4.6.15 Range input, simulated laser rangefinder.- To determine conformance to 3.7.14, the gunner's control unit shall be placed in "AUTOMATIC" and the range counter shall be observed for proper indication of 1000 meters $\pm$ 5 meters. Actuating the "LASER IN PROCESS" switch of simulated laser rangefinder of Table XI shall cause the computer to go to boresight solutions. (An indicator light may be wired into the circuit of Figure 6 to simulate this). On removal of this signal, the computer indicators shall be observed for normal solutions. Actuating the "LASER RANGE READY" switch of simulated laser rangefinder and the 2000 meter switch shall cause the gunner's range counter to move to 2000 meters $\pm$ 3 meters. All the ranges and summations of Table I shall be individually checked for conformance.

4.6.16 Range limits tests.- To determine conformance to 3.7.15, the range knobs at each control station shall be rotated and observed for conformance as specified.

4.6.17 EFC, effective full charge count tests.- To determine conformance to 3.7.16, with the computer on the ammunition specified, the EFC switch of the simulated gunner's controls shall be actuated 20 times and the EFC dial of the computer shall advance 20 counts. Allow a minimum of 2 seconds to elapse between switch actuations.

4.6.18 Automatic-manual wind switch tests.- To determine conformance to 3.7.17, the output servos shall be observed for correct solutions as wind inputs are increased from zero to 20 MPH with the wind switch in manual position. Switching to automatic shall cause the output servos to display zero wind solutions regardless of the wind input dial position. With the wind switch in automatic position, the output servos shall display no change in solutions as wind inputs are varied. These tests shall be conducted with gunner's station controls and with commander's station controls.

4.6.19 Ballistic computer solution tests.- To determine conformance to 3.7.18, the Ballistics Computer XM19 shall be subjected to the following tests using the items of test equipment specified in Table XI. Tests shall be conducted with all boresight knobs and jump knobs at their electrical center. Just prior to conducting computer solution tests, the output servo mil scales shall be checked for zero indication in boresight mode at 1000 meters. The boresight-normal switch shall bee returned to normal and "HEAT" ammunition shall be selected for all computer solution tests.

4.6.19.1 Gunner's solutions, Table II.- With computer control at the gunner's station (indicated by green light), the cant unit set at horizontal (0°), zero wind and lead input, introduce a 350 meter range at the gunner's control panel. The elevation and depression readings at the mil scale of the gunner's computer output servos shall be observed for conformance to Table II. Continue tests for all ranges and cant angles specified in Table II.

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4.6.19.2 Commander's solutions, Table III. - Testing procedures followed in 4.6.19.1 shall be repeated with the commander's simulated control set at the main gun position and with computer control transferred to the commander's station. Elevation and deflection readings at the mil scale of the commander's output servos shall be as specified.

4.6.19.3 Gunner's solutions, Table IV. - With computer control at the gunner's station, the cant unit set at horizontal (0°), and lead input not activated, introduce a 350 meter range and 20 MPH wind from left at the gunner's control panel. The deflection reading at the mil scale of the gunner's computer output servo shall be as specified in Table IV. Continue tests for all ranges, wind velocity settings and cant angle settings specified in Table II. Mil scale readings shall be as specified.

4.6.19.4 Commander's solutions, Table V. - Testing procedures followed in 4.6.19.3 shall be repeated with the commander's simulated control set at the main gun position and with computer control transferred to the commander's station. Elevation and deflection values at the mil scale of the commander's output servos shall be as specified.

4.6.19.5 Gunner's solutions, Table VI. - Place the computer unit 10543800, in vehicle position, on the powered rotary table described in Table XII. Switches for the cant unit at the gunner's and commander's controls shall be to "OFF" position. With zero wind setting, 500 meters range and computer control at the gunner's station, introduce a clockwise (to right) velocity equivalent 28.0 Mils per second into the powered rotary table. Depress and release the gunner's right lead button. Maintain traverse rate a minimum of two seconds after releasing the lead button. Mil scale readings at the gunner's output servos shall be noted within a 28 second period subsequent to the two second period after release of the lead button. The output servos shall automatically return to the zero speed solutions within 31 seconds after release of the right lead button for all tests. (Depression of the left lead button at any time shall nullify lead solutions.) Gunner's tests for all ranges and speeds specified in Table VI shall be conducted and solutions shall be as specified.

4.9.16.6 Commander's solutions, Table VI. - Testing procedures followed in 4.6.19.5 shall be repeated with the commander's simulated control set at the main gun position and with computer control transferred to the commander's station. Deflection readings at the mil scale of the commander's output servos shall be as specified.

4.6.19.7 Commander's solutions, co-ax mode, Table VII. - With the commander's simulated control set at co-ax position, the ranges specified in Table VII shall be introduced at the commander's control unit. The elevation and deflection values at the mil scales of the commander's output servos shall be as specified for each range.

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4.6.20 Illumination, range and wind indicators test. To determine conformance to 3.7.19.1, apply power to the computer system in a darkened area and observe for legibility as specified.

4.6.21 Illumination, override control and ammunition selection indicator tests.- To determine conformance to 3.7.19.2, the rheostats at the gunner's and commander's control units shall be operated and the indicators shall be observed for light intensity as specified.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and packaging data sheet MIL-P-14232/P10543900. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Ballistics Computer: XM19 is an analog computing instrument. Utilizing target range, ammunition type selection and crosswind information, it provides elevation and deflection corrections (based on firing table superelevation, drift, sight parallax, gun droop, gun jump, boresight corrections and zeroing) to a gun sighting system in the Gun, Tank: M60A1E1. It also provides corrections for vehicle cant on irregular terrain, gun tube wear and lead angle for firing against moving targets.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable packaging data sheet number (see 5.1).
(d) Applicable stock number.
(e) Order of document precedence.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1220 - A266

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COMMANDER SERVO OUTPUT

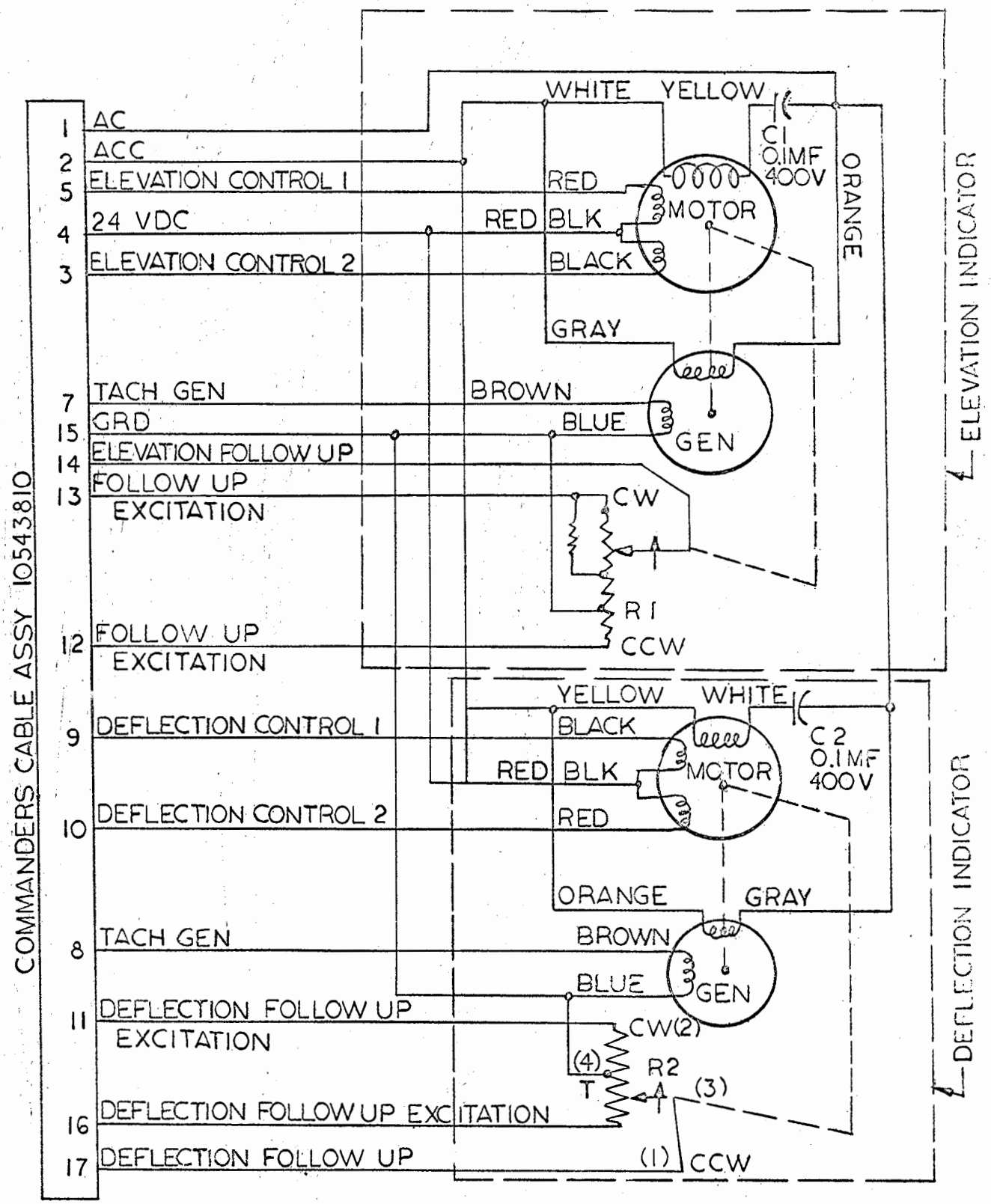


FIG. 1

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GUNNER UNIT OUTPUT INDICATORS

MIL-C-60276A (MU)

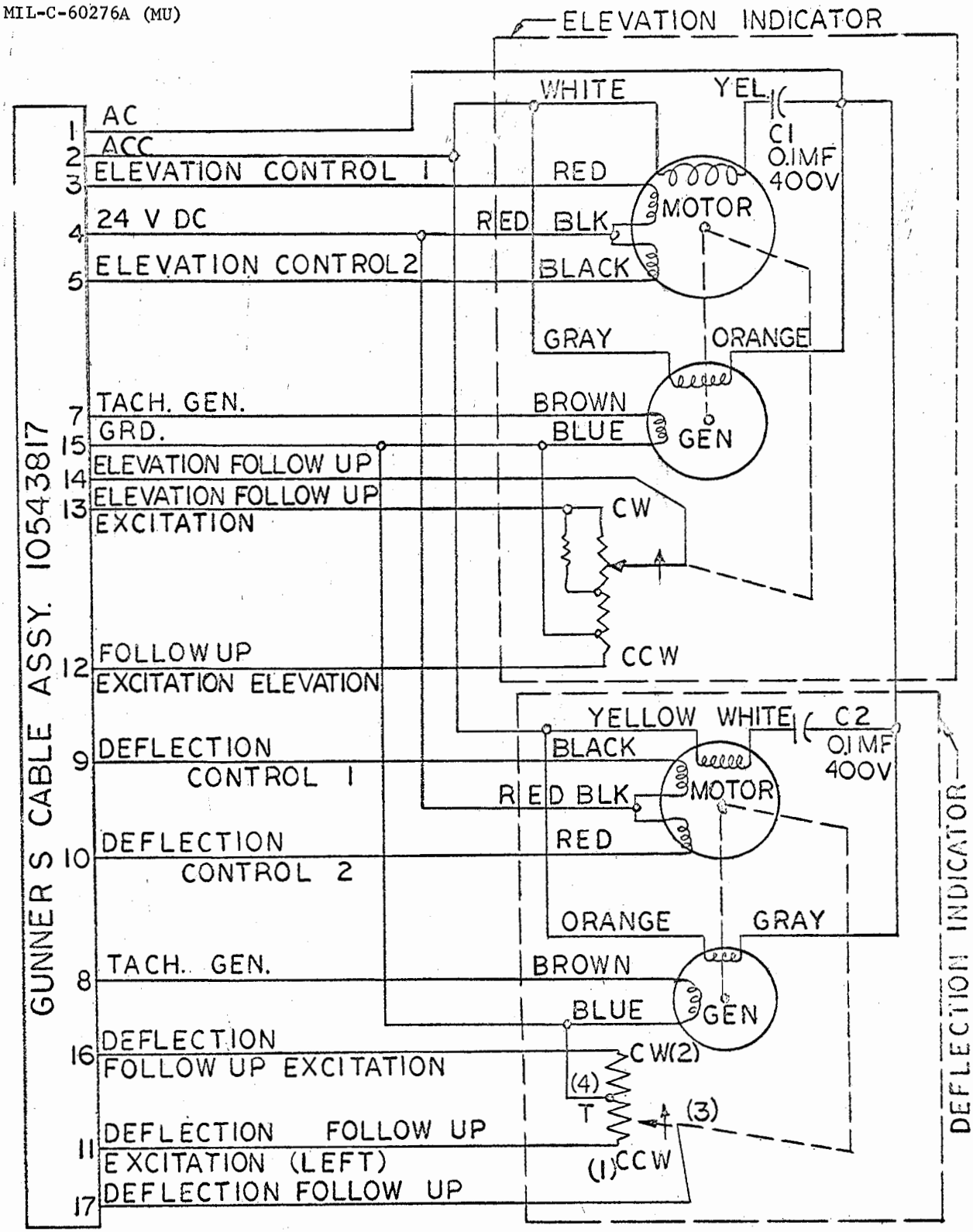
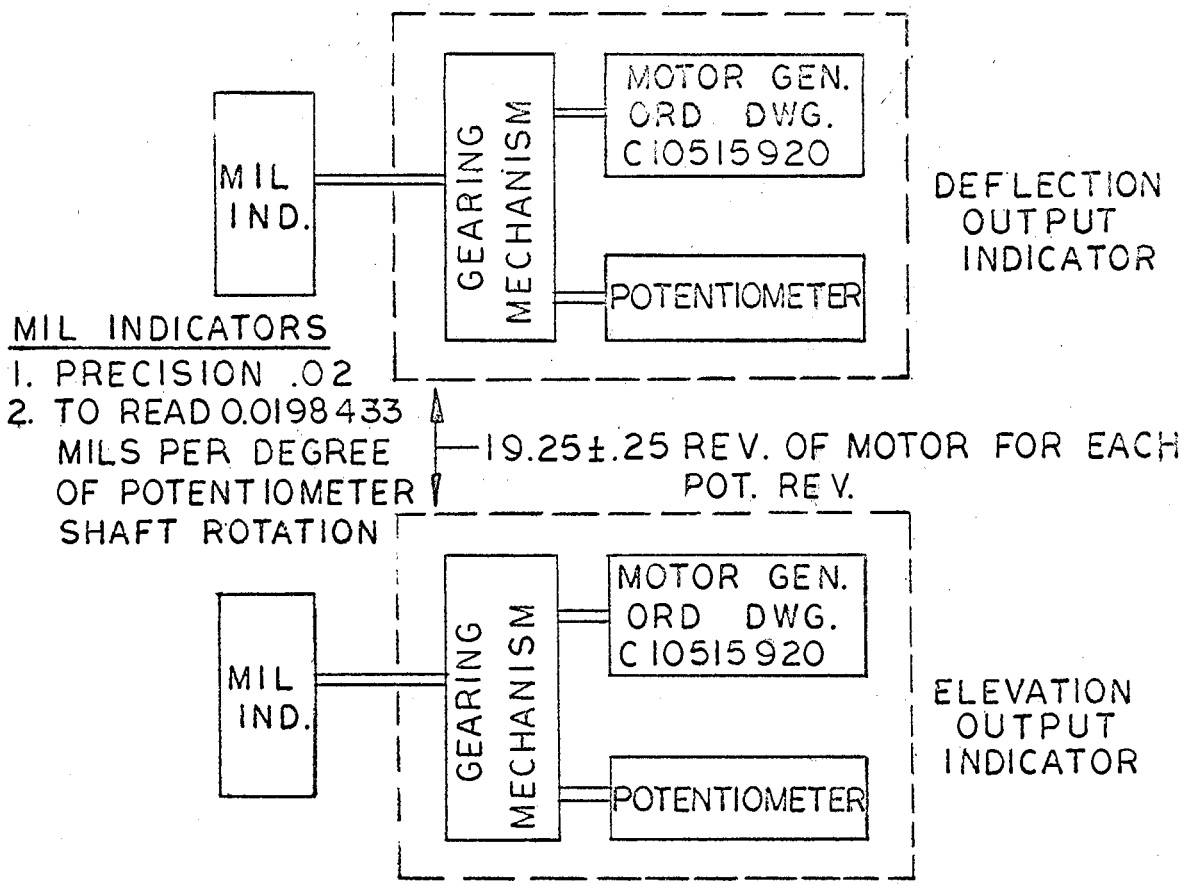


FIG. 2

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POTENTIOMETER ELECTRICAL CHARACTERISTICS

3600 DEGREES ± 1 DEGREE FROM CW TO CCW TERM.
LINEARITY ±.025%
RESISTANCE 5000 Ω ±1%

DEFLECTION POT. – CENTER TAP AT ELECTRICAL CENTER ±1°
ELEVATION POT – TAP 3174.87° FROM CCW
TAP 825.47° FROM CCW AND PAD WITH 75K ±.5%

BLOCK DIAGRAM OF SERVO OUTPUT INDICATORS FOR COMMANDER OR GUNNER

FIG 3

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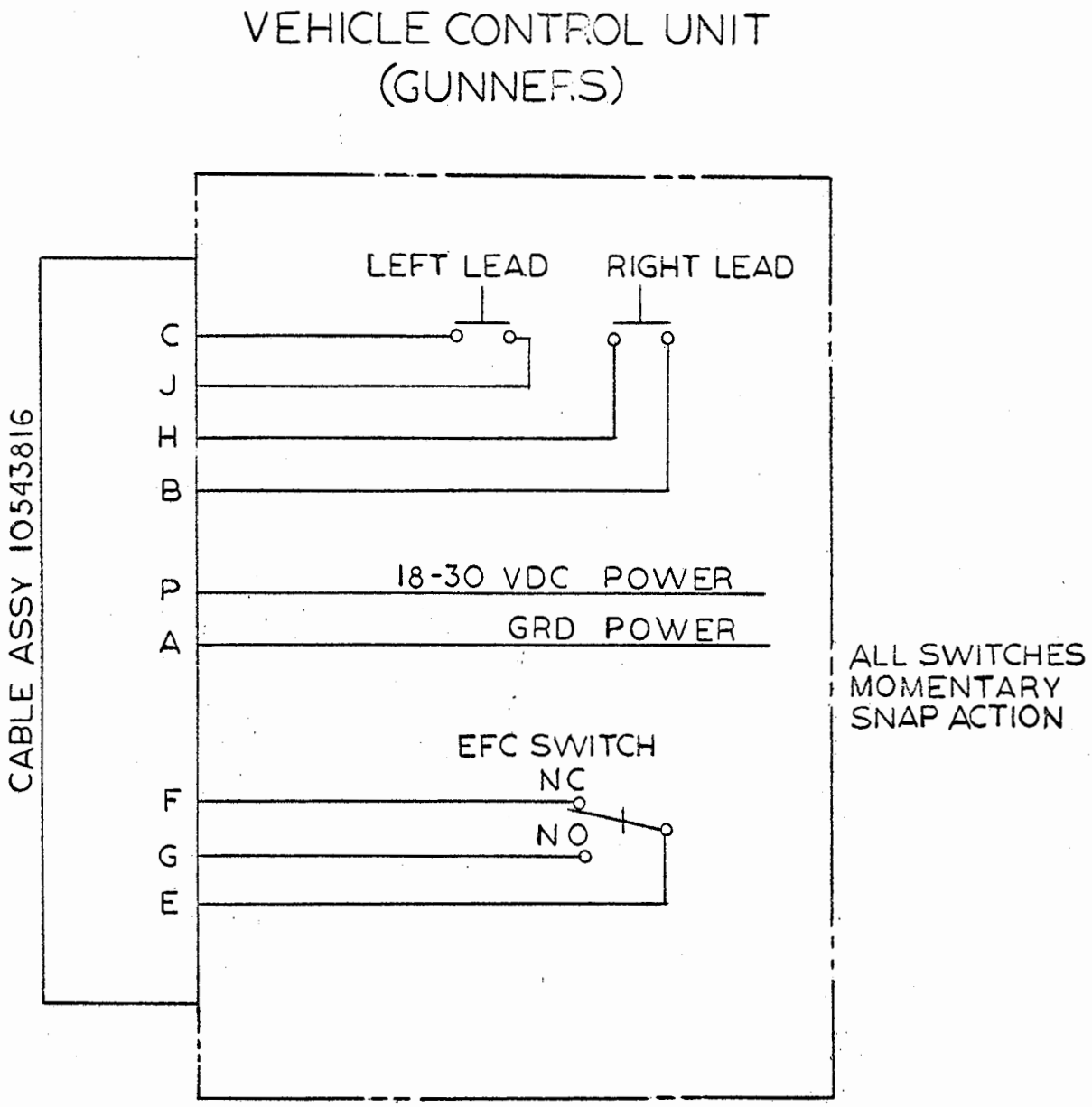
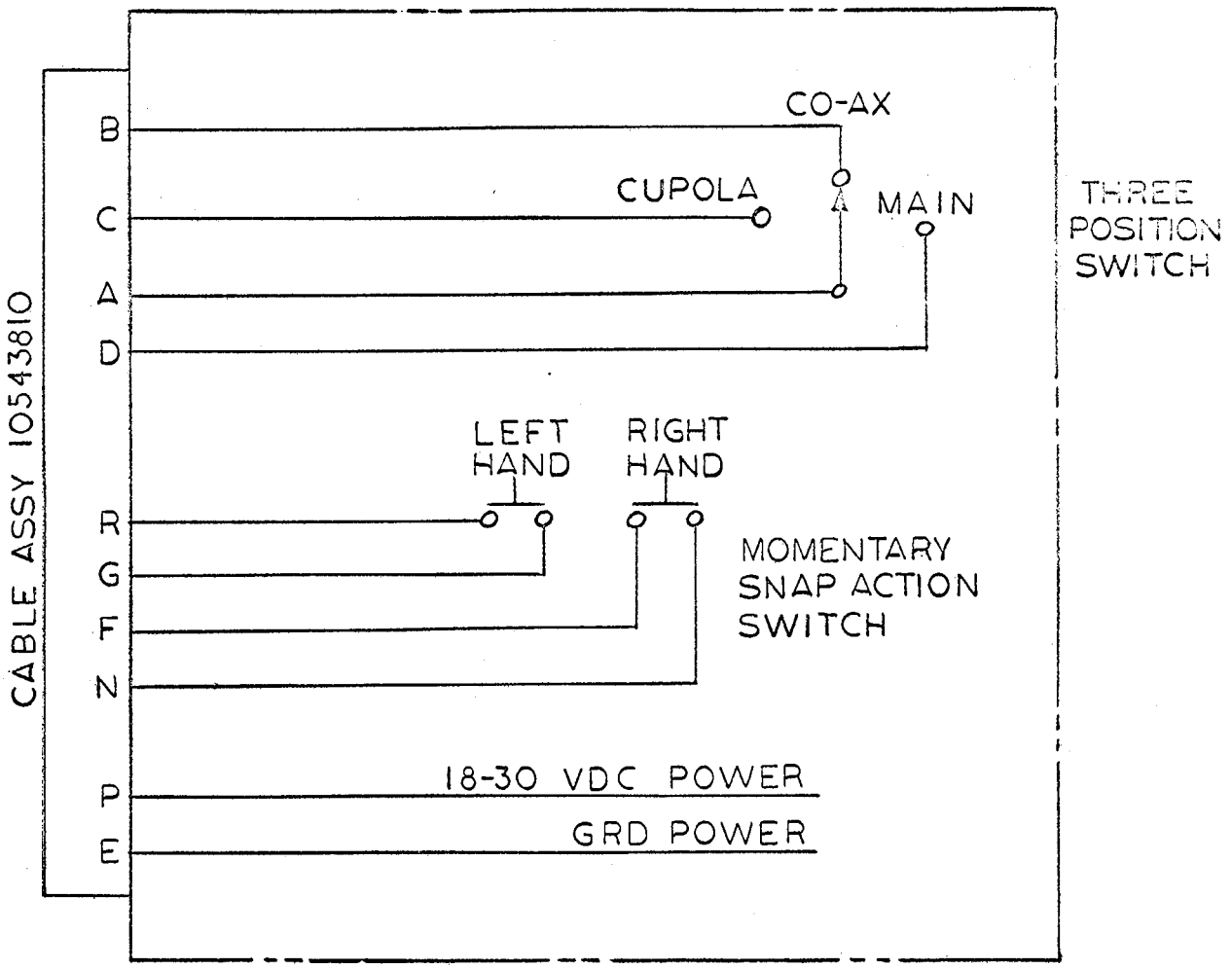


FIG. 4

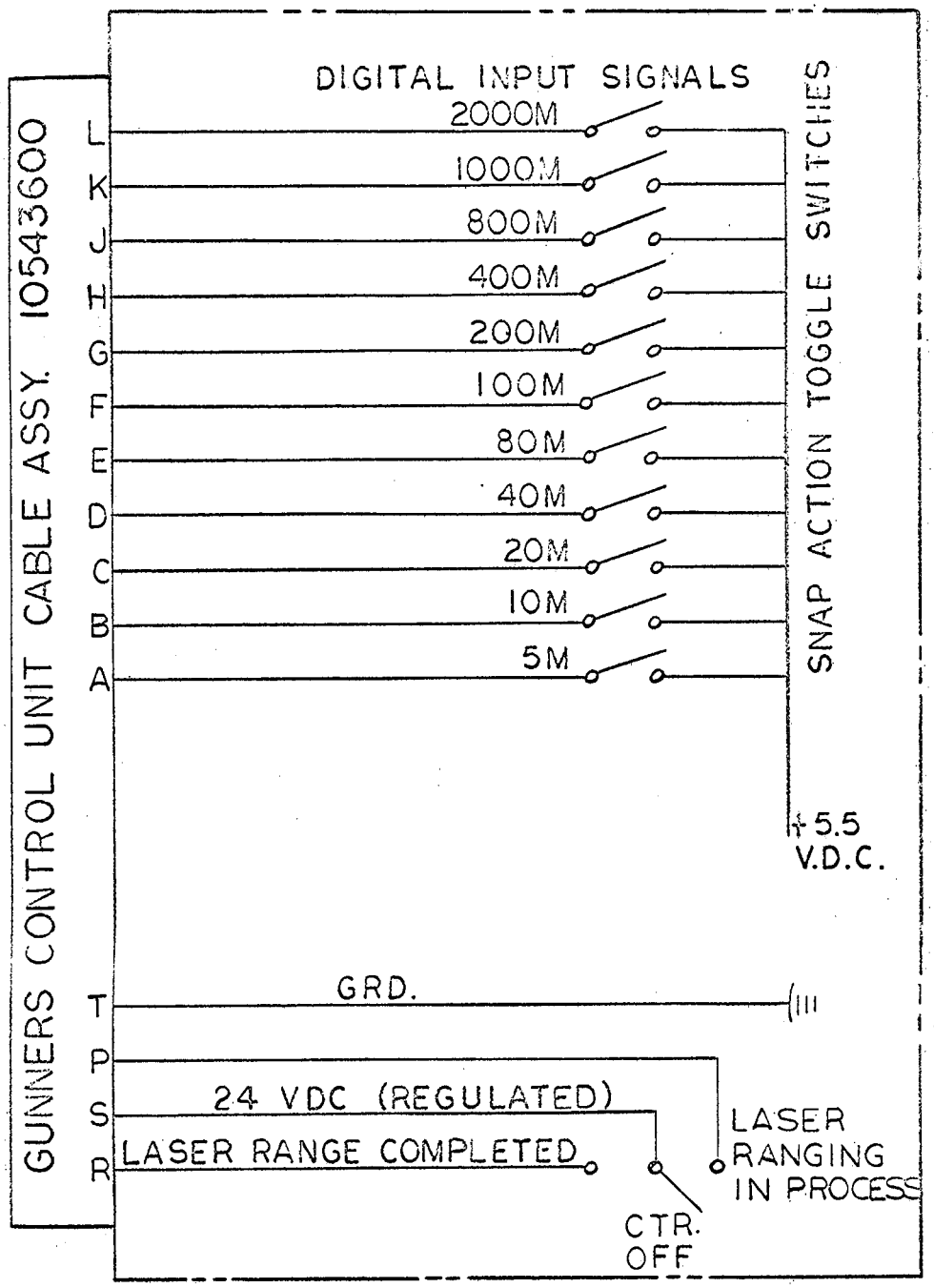
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VEHICLE CONTROL UNIT
(COMMANDERS)

FIG. 5

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SIMULATED LASER RANGEFINDER

FIG. 6

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